

The Research Problem, Questions and Objectives

90 minutes · 5 self-check questions

How to move from a broad interest to a question that is specific enough to answer, and how aims, objectives, hypotheses and conceptual frameworks fit together.

Learning objectives

- Write a problem statement that establishes magnitude, consequence and gap
- Convert a broad topic into a focused, answerable research question
- Apply FINER and PICO to test whether a question is worth asking
- Distinguish aims, objectives, research questions and hypotheses
- Draw a conceptual framework linking your key variables or concepts

The problem statement

- A problem statement answers four things in order: what is the problem, how big is it, what follows if it is not addressed, and what is not yet known
- Magnitude needs evidence — prevalence, incidence, cost, or documented reports, each cited
- The gap is the pivot of the whole proposal: it must be a gap in knowledge, not merely a gap in your own reading
- A gap can be empirical (nobody has measured this), methodological (measured badly), theoretical (no adequate explanation) or contextual (established elsewhere but never tested here)
- Contextual gaps are legitimate and are often the strongest basis for research in Ghana and across Africa

IN PRACTICE

"No study has been done in my district" is only a gap if there is reason to expect the local answer might differ. Say why it might — different health system, different exposure, different population structure.

From topic to question

- Narrow along four axes: population, concept or exposure, outcome, and setting or time
- "Malaria in children" is a topic; "Does distance to a health facility predict delayed treatment for malaria in under-fives in the Upper East Region?" is a question
- A good question is answerable with the data and time you actually have

- If you cannot name the analysis that would answer it, the question is not yet specific enough
- Quantitative questions specify variables and their relationship; qualitative questions specify experience, process or meaning

Testing the question: FINER and PICO

- FINER — Feasible, Interesting, Novel, Ethical, Relevant
- Feasibility is the most commonly overlooked: participants, time, funding, skills and access
- PICO structures comparative quantitative questions: Population, Intervention or exposure, Comparison, Outcome
- Add T for timeframe (PICOT) when duration matters
- For qualitative work, SPIDER is a better fit: Sample, Phenomenon of Interest, Design, Evaluation, Research type

Framing the same topic three ways

Element	Quantitative (PICO)	Qualitative (SPIDER)
Who	Under-fives with confirmed malaria	Caregivers of under-fives treated for malaria
What	Distance greater than 5 km to facility	Experience of seeking and delaying care
Compared with	Distance of 5 km or less	Not applicable
Outcome	Treatment delayed beyond 24 hours	How caregivers account for delay
Analysis implied	Logistic regression, odds ratio	Thematic analysis of interviews

Aims, objectives and hypotheses

- The aim is one sentence stating the overall purpose — usually beginning "To determine...", "To explore...", "To evaluate..."
- Objectives break the aim into specific, ordered, measurable steps, each starting with an action verb
- Three to five objectives is usual; more than six normally signals an unfocused study
- A hypothesis is a testable statement of an expected relationship, stated for quantitative work only
- The null hypothesis states no difference or no association; the alternative states the expected effect
- Qualitative studies state research questions rather than hypotheses, because they do not set out to test a prediction

IN PRACTICE

Objectives must map one-to-one onto your analysis plan and, later, onto your results section. If an objective produces no result, or a result answers no objective, something has gone wrong.

Conceptual and theoretical frameworks

- A theoretical framework borrows an existing theory (Health Belief Model, Theory of Planned Behaviour, Social Cognitive Theory) to explain your phenomenon
- A conceptual framework is your own diagram of how the variables in your study relate
- It should distinguish independent, dependent, mediating, moderating and confounding variables
- Drawing it early exposes variables you would otherwise forget to measure
- The framework is also what tells you which variables must be adjusted for in analysis

Key terms

Research gap

A specific limitation in existing knowledge that the proposed study will address.

Aim

A single statement of the overall purpose of the study.

Objective

A specific, measurable step that contributes to achieving the aim.

Null hypothesis

The testable statement that no difference or association exists in the population.

Conceptual framework

A diagram or model of the presumed relationships among the study variables.

Confounder

A variable associated with both exposure and outcome that can produce a spurious association.

Activity

Write a one-paragraph problem statement and one research question for your own study. Exchange with a partner, who must identify the gap type and name the analysis your question implies. If they cannot, revise it.

Further reading

- **Hulley et al. — Designing Clinical Research (Ch. 2)** — Origin of the FINER criteria; concise and practical.
- **Cooke, Smith & Booth (2012) — "Beyond PICO: The SPIDER Tool"** — The qualitative counterpart to PICO.

- **Ravitch & Riggan — Reason & Rigor** — The best treatment of conceptual frameworks as an argument rather than a diagram.

Self-check questions

1. Which of the following is a research question rather than a topic?
 - a. Adolescent mental health in Ghana
 - b. The effect of social media on young people
 - c. Is daily social media use of more than three hours associated with depressive symptoms among senior high school students in Accra?
 - d. Investigating depression and technology
2. In FINER, the criterion most often underestimated by postgraduate students is:
 - a. Interesting
 - b. Novel
 - c. Feasible
 - d. Relevant
3. Which statement is a correctly framed null hypothesis?
 - a. Distance to a health facility increases treatment delay.
 - b. There is no association between distance to a health facility and treatment delay.
 - c. Does distance to a health facility affect treatment delay?
 - d. Treatment delay is an important problem in rural districts.
4. A study on a topic well established in Europe but never examined in West Africa is justified by which type of gap?
 - a. Methodological gap
 - b. Theoretical gap
 - c. Contextual gap
 - d. No gap exists
5. The relationship between objectives and the results section should be:
 - a. Independent — results may cover whatever the data show
 - b. One-to-one — each objective yields a result and each result answers an objective
 - c. Objectives are for the proposal only and need not appear later
 - d. Results should exceed the objectives to show thoroughness

Answers

1. **c.** Only option C specifies population, exposure, outcome and setting, and it implies a definite analysis. The others name areas of interest without stating what would be measured or compared.
2. **c.** Feasibility — access to participants, time, funding and skills — is where most over-ambitious proposals fail. A question can be novel, ethical and relevant and still be impossible within a one-year programme.
3. **b.** The null hypothesis asserts the absence of an association. Option A is the alternative hypothesis, C is a question, and D is a background claim.
4. **c.** This is a contextual gap. It is legitimate, but the proposal must explain why the local answer could plausibly differ — different health system, exposure profile or population structure — rather than merely noting absence.
5. **b.** Objectives are the skeleton of the study. An objective with no result suggests work not done; a result answering no objective suggests analysis that drifted, which invites accusations of fishing for findings.